



Assessing the efficacy of China's anti-corruption drive: Insights from consumer expenditure patterns

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ABSTRACT

This paper examines the economic impact of China's President Xi Jinping's Eight Regulations of Austerity and the coordinated anti-corruption campaign. Specifically, I analyze the effects of this campaign on household expenditure patterns in China using a triple differences approach. I compare the expenditures of households with government employees to those without, both before and after the anti-corruption reform, across commodities likely to be subject to corruption versus those unlikely to be affected. The findings demonstrate that, in the post-reform period, government-affiliated families allocate relatively more expenditure towards goods they likely obtained illegally through bribes previously. Conversely, expenditure patterns for goods less susceptible to corruption do not show a relative increase in the post-reform period. These expenditure shifts are most pronounced among families of low-level officials, with the majority of the increase observed among non-Chinese Communist Party member government employees and those outside Xi Jinping's power base. This suggests that individuals with the strongest party ties were able to circumvent the effects of the anti-corruption campaign.

1. Introduction

Since the initiation of free-market reforms in 1978 in China, the Chinese government has grappled with an escalating and pervasive corruption challenge, concurrent with its rapid economic expansion and developmental trajectory (Wedeman, 2012; Dong and Torgler, 2013). Previous anti-corruption endeavors by the Chinese government have yielded relatively inconsequential outcomes when juxtaposed with Xi Jinping's anti-corruption crusade (Chen and Zhong, 2020; Ming and Liu, 2021; Ke et al., 2016, 2018). This dearth of accountability and enforcement in anti-corruption policies appears to have undergone a transformation with Xi Jinping's Eight Regulations of Austerity and coordinated anti-corruption drive commencing at the close of 2012.¹ On May 17, 2013, the Central Commission for Discipline Inspection (CCDI) inaugurated a novel anti-corruption policy.² The discernible effects of this reform are evidenced in Fig. 1, illustrating a marked escalation in the conviction rates of high-ranking government officials implicated in corruption post-2012.³

Prior to Xi Jinping's anti-corruption campaign, extant evidence suggests that a broad spectrum of Chinese government personnel,

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¹ The resolutions of the 14th National Congress of the Communist Party of China, held in January 1997, underscored the importance of anti-corruption efforts. The Central Commission for Discipline Inspection convened eight meetings dedicated to this cause. Subsequent initiatives included the formulation of the "Establishing and improving education, legal framework, punitive measures, monitoring, and corruption prevention essentials" in 2005. Additionally, the "Establishing and improving punishment and corruption prevention system 2008–2012 work plan" was developed in 2008.

² The details outlining the introduction of the Eight Regulations of Austerity can be found in Appendix page 1. These regulations primarily target the misuse of public funds by government officials for personal gain and the acceptance of bribes from private entities. In May 2013, the CCDI conducted inspections on over 270,000 bureaucrats at various levels of government.

³ In this paper, "high-ranking" refers to individuals holding provincial and ministerial-level positions. In China, the investigative process entails that governors under investigation are required to cease their official duties and reside in a detention facility.

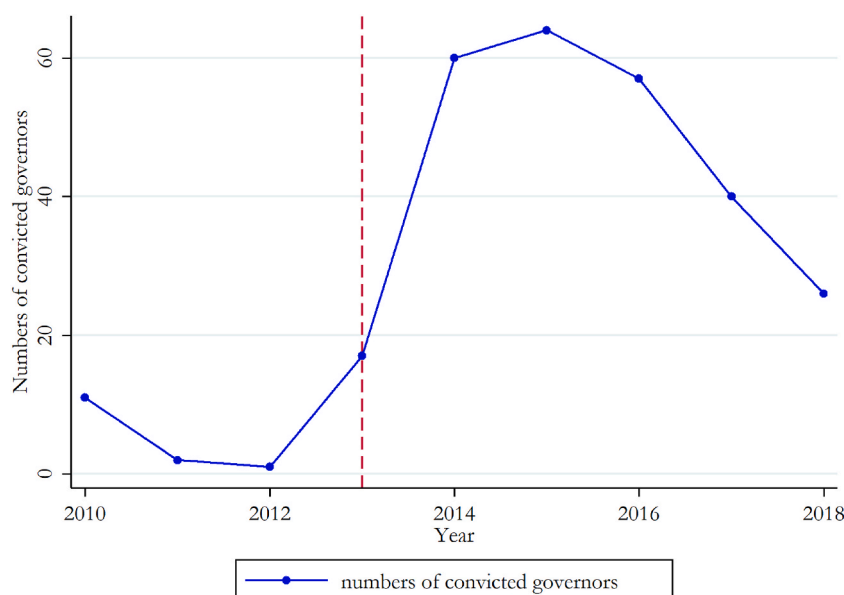


Fig. 1. Convicted Provincial and Ministerial-Level Governors

Notes: The trend depicts the number of high-level governors convicted and serving their sentence either in prison or under house detention during the investigation process from 2010 to 2018. The y-axis represents the count of convicted high-level governors.

Source: Compiled from data obtained from the Chinese government central inspection team website and author's calculations.

encompassing public servants, employees of state-owned enterprises, and institutional staff, routinely garnered illicit advantages through two primary conduits ([China Corruption Report, 2020](#)). The first and prevalent modality entails gifts, such as complimentary meals, personal utilization of government vehicles, complimentary travel, complimentary cinema and performing arts tickets, and discounts on entertainment ([Zhang, 2019](#); [Ding et al., 2020](#); [Chen and Kung, 2019](#); [Dong and Torgler, 2013](#); [Fang et al., 2018](#); [Ke et al., 2018](#); [Lin et al., 2016](#); [Qian and Wen, 2015](#); [Xu et al., 2017](#)). These gifts serve as substitutes for regular expenditures that government employees would otherwise have incurred from their own budgets. The second form of corruption involves illicit cash inflows, such as bribes from private sector entities or misappropriation of cash and other assets from public funds. Theoretically and anecdotally, higher-ranking government officials are more predisposed to receiving cash bribes, whereas lower-ranking officials are more inclined to receive non-monetary benefits, namely in-kind benefits, through bribery ([Sun, 1999](#); [T. Aidt et al., 2020](#); [Li, 2011](#)).

Xi Jinping's Eight Regulations of Austerity, along with related anti-corruption policies, were introduced to address illicit practices, with a particular focus on curbing the misuse of public funds by government officials, reducing extravagance and waste across various levels of government systems, and targeting other readily observable forms of corruption. The inspections conducted by the Central Commission for Discipline Inspection (CCDI), as part of its supervisory mechanism, assess the effective implementation of the Eight Regulations of Austerity and other coordinated anti-corruption policies. According to academic research and journalistic reports, the anti-corruption campaign specifically aimed to curb illegal subsidies and privileges enjoyed by government system workers, such as meals, vehicles, travel, and entertainment ([Zhang, 2019](#)). This initiative also resulted in reductions in expenditures on vehicles and entertainment, both financed through public funding and private bribes ([Cai et al., 2011](#); [Wang and Yan, 2020](#)). In the empirical analysis, I categorize consumption expenditures of these types as susceptible to corruption (affected expenditures), while classifying other forms of consumption expenditure as less prone to corruption (non-affected expenditures).⁴

In this paper, I employ a series of methods to investigate whether Xi Jinping's anti-corruption campaign influences household expenditure behaviors. To achieve this, I examine affected expenditures and non-affected expenditures among government and non-government households, both before and after the 2012 anti-corruption campaign. I demonstrate that in the post-reform period, households affiliated with the government exhibit a relatively higher expenditure in categories previously associated with illicit benefits, whereas expenditure patterns for items less susceptible to corruption show no significant relative increase. To gain deeper insights into the underlying shifts across different societal segments prompted by this reform, I further conduct a quintile analysis. This analysis reveals that the majority of expenditure changes occur within the lower echelons of government officials and their families. Additionally, I partition the sample into two distinct treatment groups: Chinese Communist Party (CCP) members within the government system and non-CCP members within the government system. I illustrate that only households of non-CCP government

⁴ Affected expenditures encompass categories influenced by Xi Jinping's anti-corruption campaign, including dining out, automotive expenses, other private transportation costs, public transportation, tourism, entertainment, and expenditures on durable goods. Non-affected expenditures denote categories unaffected by Xi Jinping's anti-corruption campaign, encompassing expenditures related to health enhancement, medical expenses, education, communication, clothing, daily necessities, and cosmetics.

officials experienced a compelled elevation in their expenditures on affected goods. Likewise, I observe that government officials situated in Xi Jinping's conventional powerbase provinces were least affected. This suggests that CCP members, those with the strongest governmental affiliations, may have evaded the anti-corruption campaign and persist in receiving corrupt benefits post-reform.

Recent research also underscores the impact of the anti-corruption campaign on public workers' expenditures on luxury items. Specifically, [Ke et al. \(2018\)](#) observed a significant decrease in total expenditures on luxury alcohol and upscale hotels, while expenditures on standard accommodations remained unaffected post-campaign. Similarly, [Qian and Wen \(2015\)](#) noted that the anti-corruption campaign led to a reduction in expenditures on imported luxury goods, such as jewelry. Additionally, [Xu and Yano \(2017\)](#) investigated the effects of China's Xi Jinping's anti-corruption campaign on economic growth by examining the innovation channel. They found that more private funding was allocated to research new production methods and development in companies lacking political affiliations, non-state-owned enterprises, and younger firms.

This paper makes several notable contributions to the literature on the effectiveness of anti-corruption campaigns:

First, this study categorizes households into government-affiliated and non-government-affiliated groups and analyzes household expenditures before and after the implementation of Xi Jinping's anti-corruption campaign. Leveraging data from the Chinese Family Panel Study (CFPS) dataset, the paper examines how expenditures are classified into "affected" and "non-affected" categories based on the rules and regulations underpinning the anti-corruption campaign, such as the Eight Regulations of Austerity, along with insights from academic literature, news reports, and public expenditure datasets. A comparative analysis between these two groups offers valuable insights into the campaign's effects at the household level.

Second, the study introduces an innovative research mechanism by using changes in observable expenditure behaviors to assess the effectiveness of the anti-corruption campaign on government-affiliated households' expenditures. The strength of this approach lies in its reliance on public and transparent data, avoiding the need for information on cash bribes or other clandestine corruption behaviors. This mechanism demonstrates how public data can effectively reflect the impact of anti-corruption measures.

Third, the findings reveal that the anti-corruption campaign significantly influenced consumer behavior, particularly in expenditures associated with reduced corruption. Lower-level government-affiliated households displayed increased spending on goods previously obtained at no cost or subsidized rates due to corruption. In contrast, this pattern is not evident among higher-ranking government officials. The study highlights that lower-ranked government-affiliated households allocate a significantly larger portion of their income to affected items, while the findings for higher-ranking officials remain inconclusive.

Finally, this paper provides indirect evidence suggesting that the anti-corruption campaign may have been used to eliminate political opponents. Notably, Chinese Communist Party (CCP) members and individuals residing in provinces traditionally associated with Xi Jinping's power base—those with the strongest government ties—appear to retain access to corrupt benefits even after the reforms. Additionally, government-affiliated households have shifted toward financing expenditures, previously acquired without cost, using their own income. These findings contribute to a nuanced understanding of the broader implications of the anti-corruption campaign.

2. Background on anti-corruption scheme

Corruption is widely associated with a country's development, productivity, political structure, and social equity. Traditional perspectives in the academic literature often portray corruption as a distortion that adversely affects economic development, proposing various remedies ([Shleifer and Vishny, 1993](#); [Ades and Di, 1999](#); [Avis et al., 2018](#)). Conversely, substantial evidence suggests that addressing corruption issues can bolster the country's economy while enhancing corporate innovation and productivity ([Murphy et al., 1993](#); [Veracierto, 2008](#); [Anokhin and Schulze, 2009](#); [Ellis et al., 2020](#); [Wellalage et al., 2020](#)). Many scholars have underscored corruption's potential to deter investment, exacerbate inequality, diminish effectiveness, and foster an informal economy ([Murphy et al., 1993](#); [Mauro, 1995](#); [Johnson et al., 2000](#); [Herzfeld and Weiss, 2003](#)).

Some argue that despite its drawbacks, corruption establishes informal rules that facilitate transactions, serving as either a lubricant for bureaucratic and commercial rigidity ([Lui, 1985](#); [Beck and Maher, 1986](#); [Huntington, 2006](#)) or a suboptimal allocation mechanism ([Lui, 1985](#); [Treisman, 2000](#)). Others contend that government policies and decentralized governance structures can mitigate corruption ([Fan et al., 2009](#); [Fisman and Gatti, 2002](#); [Olken and Pande, 2012](#)). Corruption exerts significant influence on political systems and societal development, particularly within the context of centralized government structures ([Levin and Satarov, 2000](#)). Some researchers have explored how anti-corruption policies in home countries can impact the business operations of international companies in regions plagued by corruption ([Karhunen and Ledyeva, 2012](#)). Certain studies have scrutinized the effectiveness of anti-corruption campaigns in other Asian nations, suggesting China's efforts are ineffective ([Quah and Yu, 2015](#)). Additionally, comparative analyses of corruption issues between the U.S. and China have been conducted ([Ramirez, 2014](#)).

Given China's political system's similarity to Russia and Russia's prior anti-corruption endeavors, it is pertinent to assess Russia's anti-corruption campaign effectiveness. Notably, Russia's anti-corruption initiatives faltered, failing to foster democratic progression ([Schulze et al., 2016](#)). [Schulze et al. \(2016\)](#) also deliberated on the influence of law enforcement and educational attainment on corrupt behaviors, asserting that higher education and robust law enforcement mechanisms can curtail corruption. These diverse research avenues render this paper distinctive, particularly as it delves into an anti-corruption campaign within a centralized government system, underscoring the importance of probing the underlying motivations behind such campaigns in countries like China.

Prior studies on Xi Jinping's anti-corruption campaign have investigated various outcomes. [Ke et al. \(2018\)](#) examined market supply-side dynamics and found that these anti-corruption measures, including the Eight Regulations of Austerity, did not affect the value of Chinese stock market shareholders. However, they did observe a negative impact on expenditures related to luxury goods and

upscale hotels. [Chen and Kung \(2019\)](#) explored the collusion between political entities and cash bribes using land transaction data spanning from 2004 to 2016. They identified local governments as the power wielders in land transactions, selling land to “businessmen” linked to China’s supreme political elites, the Politburo.⁵ These “businessmen,” often family members of political elites, reaped substantial profits, while local governors received political promotions as kickbacks. Following Xi Jinping’s anti-corruption campaign, Chen and Kung observed that the Politburo paid higher prices for land acquisitions, leading to a decrease in the promotion rate of local governors. Other studies have delved into China’s unique crony capitalism, where large private companies connected with governors initially compete with each other before merging to divide various local markets and expand their scale ([Bai et al., 2014](#)).⁶

To gain a deeper understanding of corruption behaviors in China, it is essential to first examine the structure of the Chinese government system and the hierarchical organization of its officials. The employees within the Chinese government system can be broadly categorized into three groups. The first group includes civil servants, also referred to as public servants, who work within government administrative bodies and serve as the backbone of the country’s bureaucratic system. The second group comprises employees of state-owned enterprises (SOEs), which are government-owned. The third group consists of personnel in the public service sector or public institutions, such as teachers and professors in the education system, as well as staff in cultural institutions, and news, broadcasting, and publishing units. For the purposes of this paper, these groups are collectively referred to as “government employees.” They are all part of the government employee registration system and share common benefits, such as access to superior health insurance and retirement programs. Compared to employees in other sectors, government-related employees, on average, enjoy higher incomes and additional material incentives, including consistent bonuses and subsidized groceries. Notably, individuals in higher-ranking government positions, such as local city mayors or above, are entitled to further institutionalized privileges. These disparities in compensation and material advantages reflect systemic features of China’s government system and provide an important context for understanding corruption dynamics ([Qi, 2024](#) ; [Reuters, 2024](#); [Financial Times, 2024](#); [State Council of the People’s Republic of China, 2006](#); [Ministry of Finance of the People’s Republic of China, 2013](#); [Guilin Agricultural Science Research Center, 2024](#); [State Administration for Industry and Commerce of the People’s Republic of China, 2012](#); [State Council of the People’s Republic of China, 2006](#); [National People’s Congress of the People’s Republic of China, 2018](#); [Li et al., 2024](#); [Asinta, 2024](#)).

It is widely recognized that not all Chinese government employees hold political power. Only those at certain hierarchical levels possess such authority, which allows them to wield influence and receive cash bribes. These officials are valuable targets for private companies and non-SOE entities seeking to secure favorable treatment or benefits through cash bribery. Importantly, the primary source of illicit income for higher-ranking government officials with political power often comes from cash bribes, which are inherently more difficult to detect and quantify accurately. Nevertheless, these officials must still allocate portions of this income to various expenditures, even after the implementation of Xi Jinping’s anti-corruption campaign. Consequently, the analytical framework applied in this study remain relevant for analyzing the expenditure behaviors of higher-level government-affiliated households ([People’s Daily Online, 2013](#); [Yang et al., 2024](#); [Xu et al., 2021](#)).

In contrast, many government employees without significant political power often rely on illegally obtained material benefits, which play a more critical role in supplementing their livelihoods. These illicit benefits may arise from the misappropriation of public funds, participation in private company events, or involvement in unofficial gatherings such as welcome parties. Based on the distinct characteristics of Chinese social networks, corruption behaviors in this context are also unique. Individuals seeking to build relationships with government employees often present gifts such as cigarettes and liquor. Networking with Chinese government employees frequently involves dining at grand hotels, featuring expensive meals and premium alcohol. These gatherings are typically followed by entertainment activities, such as karaoke. Consequently, dining out and entertainment, particularly karaoke, play a significant role in the analysis of corruption-related behaviors in this research paper ([Li et al., 2024](#)). However, these smaller-scale illegal material bonuses are generally of little concern to higher-ranking officials, whose primary sources of illicit income are substantial under-the-table cash bribes.

In accordance with the Eight Regulations of Austerity and the related directives of the CCDI, the central inspection teams operate under established rules and routines for conducting inspections. The impact of these inspections is more pronounced on certain specific corruption behaviors among government-related employees, such as the misuse of public funds for free meals or other unauthorized gatherings. However, their influence is comparatively limited on other forms of corruption that are more difficult for the inspection teams to detect ([Xinhua News Agency, 2023](#); [Ministry of Human Resources and Social Security of the People’s Republic of China, 2024](#)). It is also important to note that the inspection teams do not conduct comprehensive reviews of all government-related employees’ and their families’ bank accounts, nor do they require mandatory property disclosures from all government employees. This limited scope of inspection creates challenges in fully addressing and uncovering more clandestine forms of corruption. This has led to widespread speculation that Xi Jinping may have leveraged the stringent anti-corruption campaign as a means to eliminate political opponents.

It is evident that the anti-corruption campaign has influenced various forms of corruption, though some behaviors are more visibly affected than others. This distinction allows for an assessment of the campaign’s changes and effectiveness without requiring access to sensitive or confidential information, such as details of clandestine bribes. Certain forms of corruption are inherently difficult to

⁵ In this context, “businessman” denotes individuals who are family members of high-ranking government officials and who may receive various advantages through involvement in real estate ventures with local governments.

⁶ In the Chinese context, crony capitalism refers to the concentration of control over major profitable industries within the families of high-ranking government officials. This includes families of senior local governors, provincial governors, and central government officials.

conceal, making their changes more directly observable. For example, prior to the campaign, government employees might have dined out at public expense as frequently as ten times per week. During or after the campaign, this frequency may have declined to four or five times per week. This study primarily focuses on such observable changes in corruption-related behaviors, which offer concrete evidence of the campaign's impact. Conversely, less visible forms of corruption, such as cash bribes, are more challenging to analyze due to their covert nature (Pan and Chen, 2018). This research emphasizes the effects of the anti-corruption campaign on behaviors that are both easily observable and accessible for analysis.

Building on the background information presented, this paper's central analytical framework is to analyze how the anti-corruption campaign influenced changes in household expenditures directly funded by their own income. For example, prior to Xi Jinping's anti-corruption campaign, a household's total expenditure on dining out might have been \$1,000, with \$200 sourced from the household's own income and \$800 derived from in-kind bribes. This in-kind benefit might reflect government employees misappropriating \$800 of public funds to finance their extravagant meals. Following the campaign, the total expenditure on dining out could decrease to \$800, with \$300 now coming from the household's own income and \$500 from remaining in-kind bribes. This study investigates and compares changes in both affected and non-affected expenditure categories funded by households' own income before and after the anti-corruption campaign. Specifically, the analysis focuses on whether household expenditure on dining out, funded by their own income, increased from \$200 to \$300, illustrating the broader impact of the anti-corruption campaign on expenditure behaviors. The results indicate that only lower government-affiliated households experienced an increase in affected expenditures, likely funded by their own legal income due to their limited political power and access to high-value bribes. This inference aligns with the broader analysis of expenditure patterns in the context of the anti-corruption campaign. Similar to findings in other studies, which observed decreases in total expenditures on specific goods, this example illustrates a reduction in total dining-out expenses, declining from \$1000 to \$800. This focus offers critical insights into the broader implications of anti-corruption measures on household expenditure behavior. Future access to more comprehensive expenditure data, including in-kind bribes, would enable deeper analysis of changes in in-kind benefits and provide a more nuanced understanding of the broader impacts of the anti-corruption campaign.

Considering the structure of the Chinese government system, the distinctive nature of Xi Jinping's anti-corruption campaign, and the research framework, this paper investigates how the campaign has influenced observable expenditure behaviors. Specifically, it examines changes in expenditure categories classified as "affected expenditures" and "non-affected expenditures" among government-related households. To guide this analysis, I propose the following two general hypotheses regarding the effectiveness and implications of the anti-corruption campaign:

Hypothesis 1. The anti-corruption campaign has led to significant changes in certain expenditure patterns among government-related households.

Hypothesis 2. Indirect evidence derived from public expenditure data may support the anecdotal claim that Xi Jinping is leveraging the anti-corruption campaign to target his political rivals.

In China, Xi Jinping's vigorous anti-corruption campaign, launched on December 4th, 2012, has reshaped the political landscape and economic structure. Despite the existence of anti-corruption policies preceding Xi Jinping's campaign, their impact was minimal, as depicted in Fig. 1. Several factors contributed to the ineffectiveness of previous anti-corruption initiatives, including a weak inspection system, lengthy intervals between inspections, lack of internal oversight in monitoring, a convoluted reporting mechanism, and the absence of high-profile prosecutions of senior officials.

To enhance the efficiency of the anti-corruption campaign, Xi Jinping's administration implemented several reforms within the CCDI inspection framework and intensified inspection frequency. Moreover, the campaign sought to refine its inspection targets, shifting from broad and generalized objectives to more specific and actionable criteria. For example, the focus included detailed audits of local government accounting records to identify expenditures on unwarranted public banquets and discreetly monitoring whether government employees frequently visited upscale hotels for "free meals.". Another innovation involved frequent rotation of leaders within inspection teams, a measure designed to prevent the formation of close ties between inspection teams and the departments under scrutiny. Additionally, every monitoring activity now necessitates confirmation from the Central Committee of the Communist Party of China. A significant development was the introduction of a user-friendly online reporting system accessible to all individuals, regardless of their affiliation with the Chinese government. Finally, the campaign underscored its commitment to transparency and accountability through the public prosecution of high-ranking officials, including prominent figures like Xilai Bo, Member of the Central Political Bureau; Zhou Yongkang, a member of the Politburo Standing Committee and Secretary of the Central Committee of Politics and Law; Boxiong Guo, a member of the Politburo and vice president of the Central Military Commission; and Caihou Xu, Vice President of the Central Military Commission.

3. Data set

To investigate this topic, I utilized data from the Chinese Family Panel Study (CFPS) spanning from 2010 to 2018. According to the CFPS user guide, the dataset encompasses 25 provinces, municipalities, and autonomous regions in China, excluding Hong Kong, Macao, Xinjiang, Tibet, Qinghai, Inner Mongolia, Ningxia, and Hainan. The population covered by these 25 regions accounts for approximately 95 % of China's total population. Therefore, the CFPS can be considered a nationally representative sample. The CFPS is designed to document China's historical changes, providing an invaluable resource for both current and future social science research. It recognizes that social phenomena span multiple interconnected domains, such as family background, education, expenditures, and health, which are deeply interdependent. These phenomena also operate across different levels, from macroscopic factors like state

policies and cultural traditions to intermediate influences such as regional economies and community environments, and finally to microscopic factors including family structures and individual careers. Furthermore, the CFPS emphasizes the temporal continuity of social phenomena, acknowledging that present behaviors are shaped by past events and future decisions are influenced by current experiences. This design highlights the complexity and dynamic nature of social life in China.

The CFPS was not explicitly designed to examine the effects of the anti-corruption campaign, as it does not include survey questions directly addressing corruption. Instead, its expenditure-related questions focus on routine household expenses, capturing data on normal life expenditures. For instance, one question asks, “Not including being treated, how much money on average does your family spend on eating out per month?” Responses to such questions are likely more reliable, as participants have little motivation to misrepresent their answers. The CFPS survey utilized face-to-face interviews aided by Computer-Assisted Personal Interviewing (CAPI) technology and incorporated Computer-Assisted Telephone Interviewing (CATI). These technologies were implemented to enhance both the efficiency and quality of the survey process. Designed by a research team at Peking University, the CFPS is implemented by the (Institute of Social Science Survey (ISSS), 2015) with support from Peking University, the National Natural Science Foundation of China, and various government agencies. The University of Michigan also contributed to the survey design. The CFPS is a comprehensive longitudinal panel dataset. At the household level, the survey reported a response rate of 81.25 %, a cooperation rate of 96.58 %, a contact rate of 84.13 %, and a refusal rate of 2.67 %. At the individual level, the corresponding rates were 84.14 %, 87.01 %, 96.7 %, and 8.47 %, respectively.

This dataset provides detailed information on family-level expenditures and individual demographic characteristics.⁷ Moreover, the dataset incorporates sample collection weights tailored to different provinces, accounting for unique provincial characteristics such as population size, environmental factors, and economic conditions. Given that the 2010 wave lacks detailed expenditure categories, I focused my analysis on data from 2012 to 2018 for primary regression analysis and several robustness checks. Nonetheless, I also included the 2010 dataset in some robustness tests, leveraging its information on annual total family expenditure and government membership, consistent with subsequent waves.⁸ In the future, I aim to obtain expenditure data of similar categories spanning longer time periods prior to the anti-corruption campaign to strengthen the robustness of this research. Additionally, acquiring more detailed and granular expenditure information would provide an opportunity to conduct a deeper and more comprehensive investigation into this topic. While the dataset covers specific expenditure categories, it does not encompass the all kinds of expenditures. Notably, certain expenditure items lack specificity, hindering the identification of substitution elasticities between expenditure categories. For instance, car expenses and other transportation costs are amalgamated, with “other transportation expenses” potentially encompassing motorcycles and bicycles. Prior to regression analysis, I examined the income elasticity of demand for subcategories within affected and non-affected expenditures, as presented in Table 2. A positive income elasticity indicates that as total family income increases, expenditures within these categories also rise. Additionally, Fig. 3 illustrates the average total family income for government and non-government households, revealing a parallel trend over the analyzed period.

For enhanced precision in identification, this study scrutinizes household-level expenditures, as individual expenditures may be influenced by the spending patterns of other family members. The expenditure data in the CFPS dataset does not focus on specific goods but instead on broader expenditure categories. This approach ensures that expenditure shifts—such as substituting high-cost goods for more affordable alternatives—remain captured within the same expenditure category, simplifying the estimation of changes. For example, survey questions include: “In the past 12 months, the total expenditure on entertainment (including purchasing books, newspapers, magazines, VCDs, and DVDs, and going to cinemas and bars, and so on) was _ yuan” and “In the past 12 months, the total expenditure on tourism (including the costs of transportation, accommodation, catering, guide service, entrance tickets, and so on) was _ yuan.”

All expenditure-related survey questions were consistently administered across all waves of the CFPS survey, with the exception of the first wave, which only included total household expenditure data. This consistency across subsequent waves ensures the reliability and comparability of the data. The approach of tracking expenditure changes in this study does not require that government-affiliated households entirely abstained from affected or non-affected expenditures before the anti-corruption campaign and began such expenditures only after its implementation. To clarify this approach further, consider the following example: prior to the anti-corruption campaign, government-affiliated households spent approximately \$400 per month on dining out. After the campaign, this amount increased to approximately \$800 per month. This example illustrates that, prior to the anti-corruption campaign, government-affiliated households were able to engage in activities such as dining out with minimal reliance on their personal finances, as some of these expenses were likely subsidized through corrupt practices. Following the implementation of the campaign, however, these households may have needed to allocate a larger proportion of their own income to maintain the same level of dining out. While the overall expenditure on dining out may have decreased after the anti-corruption campaign compared to the total expenditure (including previously subsidized activities) prior to the campaign, the proportion of dining expenditures funded directly from personal income likely increased. This shift primarily reflects a change in the source of funding rather than a complete elimination of such expenditures. Based on the findings of this research, it is highly probable that the increase in expenditures was financed through legal income sources. Employing existing expenditure data from the CFPS dataset, alongside insights from prior research and news reports, different expenditure categories are categorized into affected and non-affected expenditure groups (Li et al., 2024; Chen and Zhong, 2020; Chen

⁷ For further information about the CFPS, readers can refer to the users' handbook available online (Xie et al., 2017) “Onsite sampling frame building in CFPS 2010 baseline survey” (in Chinese), Technical Report CFPS-2, Institute of Social Science Survey, Peking University, Beijing. The handbook can be accessed at: <https://opendata.pku.edu.cn/dataverse/CFPS>.

⁸ In addition, information from 2010 was incorporated for placebo tests and common trend checks.

and Kung, 2019; Dong and Torgler, 2013; Fang et al., 2018; Ke et al., 2018; Lin et al., 2016; Qian and Wen, 2015; Xu and Yano, 2017).⁹ For instance, Xun Li's research, which also utilizes the CFPS dataset, highlights the impact of the anti-corruption campaign on dining-out behaviors, noting its contribution to improved health indices in China. The study further discusses the decline in alcohol-related gift bribes as a significant outcome of the campaign. Another example, as highlighted in a report by the *South China Morning Post*, focuses on the areas targeted by anti-corruption efforts, such as expenditures on meals, travel, stationery, and office supplies, as detailed in *People's Daily* (Zhang, 2019). Lower-level civil servants have reported feeling the impact of these measures, with reduced organizational activities such as outings and meals, as documented in Wang's account (Zhang, 2019). The classification of affected and non-affected expenditures is based on the criterion of focusing on observable material forms of illegal corruption. These tangible forms of corruption are more accessible for analysis using available data and offer valuable insights into the effectiveness of the anti-corruption campaign. Furthermore, they may provide critical evidence regarding the underlying objectives of the campaign. Therefore, according to the previous research results, the CFPS data sets expenditure information, and the characteristics of Chinese government system, and Xi Jinping's anti-corruption uniqueness, I classify the eating out, car expenditure, other transportation, public transportation, travel, entertainment, durable goods as affected expenditure and health expenditure, medical expenditure, education expenditure, communication expenditure, cloth expenditure, necessities, and cosmetic as non-affected expenditure. In China, the health insurance system for Chinese government employees is much better than the non-government employees. Additionally, the hospital is part of the Chinese government system, hospital would automatically provide better health care and services for government employees, especially for higher level government employees. For example, in Beijing 301 hospital is specialized for top Chinese governors such as Xi Jinping. It is similar for education resources. Chinese government employees' children, especially the higher-level government employees can legally access better education resources (Fang, 2011; Wei and Song, 2024). For items such as clothing and cosmetics, it is common for bribers to offer gift cards or cash to the bribe recipients, as they may not be familiar with the recipient's specific preferences. More importantly, due to the uniqueness of Chinese culture, people prefer dining out, entertainment, and sending gifts as alcohol and cigarettes.

Furthermore, in the robustness section, the study examines the sensitivity of its findings to the composition of goods selected for inclusion in the affected and non-affected expenditure groups.

Households containing at least one government member are classified as government households, while those lacking government affiliation are categorized as non-government households. In regression analyses, family-level expenditure serves as the dependent variable, with demographic characteristics of the household head and other household-level information utilized as control variables.

Table 1 provides a summary of the variables utilized in the regression analysis, capturing demographic and occupational characteristics at both the individual level (household head) and the household level. The table is organized into four columns: column (1) presents summary statistics for the entire dataset ("Full"), column (2) for government-affiliated households, column (3) for non-government-affiliated households, and column (4) for the *t*-test results, indicating whether the differences between columns (2) and (3) are statistically significant.

The distinction between individual-level and household-level information arises because certain variables are inherently more relevant to one level than the other. For instance, occupational characteristics of the household head are presented in Panel A, titled "Characteristics of the Household Head's Occupation." Within the Chinese government system, employment can be categorized into three types: public servants (civil servants) operating within administrative structures, employees of state-owned enterprises (SOEs), and individuals in public institutions such as hospitals, schools, and research centers. For comparison, "Other" includes household heads employed in private companies, international firms in China, self-employed roles, and similar occupations outside the government system.

The full dataset, represented in column (1), indicates that the proportion of household heads employed as public servants is 0.02, while the rates for those employed in SOEs and public institutions are 0.04 and 0.03, respectively. The number of households in column (2) (government-affiliated) is 10,120, while column (3) (non-government-affiliated) includes 52,725 households. Column (2) reveals that among government-affiliated households, 11 % of household heads are public servants, 26 % are employed in SOEs, and 16 % are in public institutions. Interestingly, 46 % of household heads in government-affiliated households fall under the "Other" category, indicating that while these households are government-affiliated, the household head may not work within the government system—suggesting other family members might hold government positions.

In contrast, column (3) shows that all household heads in non-government-affiliated households are categorized under "Other," as these households lack any members employed in the government system. The *t*-test results in column (4) indicate statistically significant differences between the occupational distribution of household heads in government-affiliated versus non-government-affiliated households. For example, the proportion of government-affiliated household heads working in non-government jobs is significantly lower (0.54) than that of non-government-affiliated household heads, with the difference being significant at the 0.01 level.

This analysis highlights the occupational composition of household heads across different household categories, providing essential context for understanding variations in household characteristics within the dataset. The distinctions between government-affiliated and non-government-affiliated households underscore the nuanced relationship between household roles and occupational classifications in the broader context of this study.

In Panel B of Table 1, titled "Educational Attainment of the Household Head," the data reveals a statistically significant higher level

⁹ More details on the classification method are available in the Appendix.

Table 1
Summary statistics of demographic characteristics and household attributes.

	(1)	(2)	(3)	(4)
	Full	Government	Non-Government	Difference
A: Characteristics of the Household Head's Occupation				
Public servant	0.02	0.11	0	0.11
SOE	0.04	0.26	0	0.26
Institution	0.03	0.16	0	0.16
Other	0.91	0.46	1	−0.54***
B: Educational Attainment of the Household Head				
< Primary	0.26	0.10	0.28	−0.18***
Primary	0.21	0.10	0.23	−0.12***
Middle school	0.28	0.25	0.28	−0.03
High school	0.14	0.24	0.12	0.11***
Junior college	0.06	0.15	0.04	0.11***
Undergraduate	0.04	0.13	0.02	0.11***
Graduate	0.00	0.01	0.00	0.01***
C: Basic Demographic Information				
Not Han	0.03	0.01	0.03	−0.01***
Urban	0.48	0.72	0.44	0.28***
Hukou	0.29	0.60	0.23	0.35***
Female	0.50	0.50	0.50	−0.006
Family Size	3.34	3.51	3.31	0.19***
	(1.63)	(1.53)	(1.65)	(0.02)
Adult Size	2.52	2.79	2.46	0.34***
	(1.20)	(1.24)	(1.19)	(0.0153)
Age	46.73	42.15	47.69	−5.72***
	(16.94)	(14.50)	(17.26)	(0.192)
D: Regional Attributes				
East	0.43	0.45	0.43	0.0233***
Middle	0.29	0.32	0.29	0.03***
West	0.27	0.23	0.28	−0.05***
N	62,845	10,120	52,725	

Note: The sample includes 62,845 observations from 2012, 2014, 2016, and 2018, with 52,725 from non-government households and 10,120 from government households. Panel A, “Characteristics of the Household Head’s Occupation,” summarizes individual-level data on household heads: “Public servant” refers to government administrative roles, “SOE” denotes State-Owned Enterprises, and “Institution” includes government-affiliated research, education, and welfare roles. Panel B, titled “Educational Attainment of the Household Head,” presents individual-level data. The category “<Primary” refers to less than primary education, “Primary” corresponds to primary school, while “Middle School” and “High School” align with secondary education levels in the United States. The “Graduate” category encompasses advanced degrees, including master’s and doctoral degrees. Panel C, titled “Basic Demographic Information,” includes both individual-level data (e.g., “Female,” “Age”) and household-level data. The category “Not Han” represents ethnic minority groups, while “Urban” indicates the proportion of households residing in urban areas. “Hukou” reflects urban registration rates, “Family size” is the total household members, and “Adult size” is the number of adults. Panel D, “Regional Attributes,” contains household-level data: “East,” “Middle,” and “West” denote geographical regions in China. Column (4), “Difference,” reports *t*-test results for differences between government and non-government groups. Statistical significance levels are ****p* < 0.01, ***p* < 0.05, and **p* < 0.1.

of education among household heads in families with government employees, with the exception of middle school education. This suggests that government-affiliated households are more likely to have heads with higher educational qualifications compared to their non-government counterparts.

In Panel C, titled “Basic Demographic Information,” several noteworthy distinctions emerge between government-affiliated and non-government-affiliated households. The variable “Not Han” represents ethnicity at the household level, encompassing all 55 recognized ethnic groups in China except the Han ethnicity. The proportion of “Not Han” ethnicity is higher among non-government-affiliated households compared to those within the government system.¹⁰ Additional household-level variables include “Urban,” “Hukou,” “Family Size,” and “Adult Size,” while “Female” and “Age” pertain to individual characteristics of the household head. The data highlights that a greater proportion of government-affiliated households reside in urban areas, where they benefit from improved access to basic utilities and social welfare services. Additionally, household heads in government-affiliated families are, on average, five years younger than those in non-government-affiliated households. Despite these differences, the proportion of female household heads is similar across both groups. Furthermore, government-affiliated families tend to have larger family sizes and a greater number of adults compared to their non-government-affiliated counterparts. These demographic patterns provide valuable context for understanding the structural differences between government-affiliated and non-government-affiliated households in this study.¹¹

China exhibits notable regional economic disparities, with distinct levels of development across its eastern, central, and western

¹⁰ In China there are 56 ethnicities. Han was the major part which is 91.11 % in 2021(Ning, 2021).

¹¹ Adult size is the number of adults in a household.

Table 2

Summary statistics of household expenditures, income, and savings.

	(1)	(2)	(3)	(4)	(5)
	Full	Government	Non-Government	Difference	Income Elasticity
A: Affected Expenditure					
Affected Expenditure	11,797	20,294	10,038	10,256.5***	
Eating Out	2679	4704	2260	2444.3***	0.38
Car	2738	4800	2342	2458.5***	0.33
Other Transportation	914	1268	840	427.6***	0.26
Public Transportation	2086	3206	1870	1335.5***	0.25
Travel	904	2068	663	1404.4***	0.28
Entertainment	261	540	208	332.4***	0.24
Durable Goods	1591	3021	1295	1725.2***	0.23
B: Non-Affected Expenditure					
Non-affected Expenditure	12,998	17,057	12,158	4899.7***	
Health	277	448	241	206.7***	0.141
Medical	4069	4122	4058	63.66	0.064
Education Expenditure	3163	4676	2872	1804.2***	0.168
Communication	1677	2380	1542	838.0***	0.143
Cloth	1967	3268	1717	1550.7***	0.225
Necessities	827	1070	777	293.5***	0.114
Cosmetic	423	785	348	436.4***	0.393
C: Pooled Family Surveyed Total Values					
Total Expenditure	35,609	51,038	32,647	18,391.4***	
Income	44,069	66,961	39,578	27,382.4***	
Saving	28,983	54,505	24,111	30,394.1***	
N	62,845	10,120	52,725		

Notes: The pooled sample encompasses data from 2012, 2014, 2016, and 2018, with all values adjusted to 2010 RMB prices using the 2010 Consumer Price Index (CPI). “Eating Out” refers to expenses incurred at restaurants. “Car” expenditure includes costs associated with vehicle ownership, including mortgages. “Other Transportation” encompasses expenses related to alternative modes of transportation, such as bikes. “Public Transportation” denotes expenditures on public transit as a substitute for private vehicle usage. “Travel” represents annual expenditures on travel-related activities. “Entertainment” covers expenses related to recreational activities such as movies and karaoke. “Durable Goods” includes significant purchases of electrical appliances, such as TVs and refrigerators. “Affected Expenditures” comprise the sum of spending on eating out, cars, other transportation, public transportation, travel, entertainment, and durable goods. “Health” denotes expenditures on fitness or nutritional supplements. “Medical” encompasses all household expenditures related to medical treatments. “Education Expenditure” represents household spending on educational pursuits. “Communication” includes expenses associated with postage and phone call services. “Necessities” refers to daily expenditures excluding all of the aforementioned categories. “Cosmetic” represents household spending on beauty products or treatments. The total number of observations is 62,845, with 10,120 for government households and 52,725 for non-government households. Column (4), labeled “Difference,” displays the results of a *t*-test comparing the government and non-government groups. Statistical significance levels are denoted as ****p* < 0.01, ***p* < 0.05, and **p* < 0.1.

regions. Major cities such as Beijing, Shanghai, Jiangsu, and Guangdong characterize the comparatively affluent eastern part, while the central region, encompassing provinces like Anhui, Sichuan, and Henan, remains less developed. The western region represents the most economically underdeveloped areas. To mitigate the influence of time-invariant unobservable regional discrepancies, regional dummies are included in all regression analyses.

In addition to basic demographic information, the CFPS also provides data on annual total family incomes, expenditures, and expenditure categories, as detailed in Table 2. Household expenditures are presented in real terms, adjusted to 2010 consumer prices. I categorize all expenditures into two groups: those that are easily and commonly open to corruption, labeled as “affected expenditures” in panel A, and those less susceptible to graft, referred to as “non-affected expenditures” in panel B. “Affected” expenditures are those expected to change for households that previously received gifts substituting for real consumption in these categories. The classification relies on previous studies, newspaper reports, and CFPS expenditure information (Lin et al., 2016; Qian and Wen, 2015; Cai et al., 2011; Ke et al., 2018; Chen and Zhong, 2020; Zhang, 2019).

From Table 2, the “affected expenditure” group includes costs related to eating out, car expenditures, transportation, tourism, entertainment, and durable goods. On average, government households report affected real expenditures of 20,294 RMB (approximately 3051 USD at the 2020 exchange rate), which is significantly higher than those of non-government households. “Non-affected expenditures” include health, medical, education, communication, clothing, other necessities, and cosmetic expenditures. Although non-affected consumption expenditures are also significantly higher in government households compared to non-government households, the magnitude of this difference is smaller. Panel C indicates that annual total family expenditures in government households amount to 51,038 RMB (around 7673 USD), significantly higher than the 32,647 RMB (around 4908 USD) reported by non-government families.

Table 2 illustrates that all government household expenditures are significantly higher than those of non-government households. Additionally, government households exhibit higher annual family incomes and savings compared to their non-government counterparts. Thus, on average, government households display better economic conditions and higher expenditures than non-government families. These observable differences in economic conditions are controlled for in the analysis to ensure they do not confound the

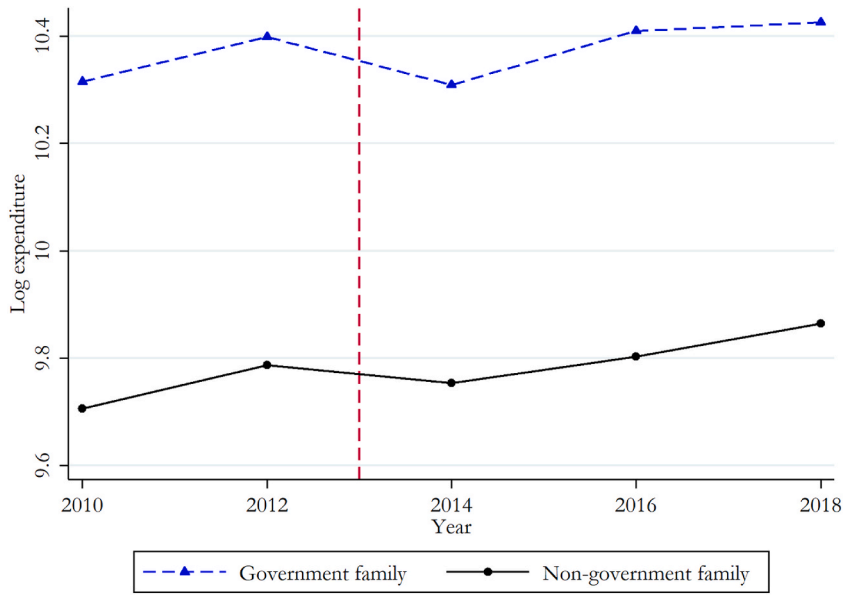


Fig. 2. Trend in Surveyed Total Family Expenditure

Notes: The data represents the Chinese Family Panel Study (CFPS) survey of family annual total expenditures spanning for the years 2010, 2012, 2014, 2016, and 2018.

Source: Author's calculations.

results, as identification is based on variations within household types over time across different sets of goods.

Besides observable factors, unobservable factors such as spending habits and behavioral trends also pose concerns. To address this, Fig. 2 depicts the annual expenditure patterns of government and non-government households from 2010 to 2018. The figure shows similar trends in total annual family expenditures for both groups, alleviating concerns about picking up differential income paths.

4. Empirical approach

4.1. Regression model

Estimating the relationship between anti-corruption policies and households' expenditures encounters a critical challenge: the likelihood of expenditure changes among government-affiliated households being correlated with variables such as age, education levels, or other unobservable factors. To address this potential endogeneity issue, a difference-in-differences approach is employed, enabling a comparative analysis across different expenditure categories within households.

Moreover, the unexpected nature of Xi Jinping's anti-corruption campaign poses additional complexities. Following the 18th National Congress of the Communist Party of China, which primarily deliberated economic conditions and political reform, Xi Jinping's swift announcement of the aggressive anti-corruption campaign just eighteen days later, upon assuming the presidency, caught many economic agents off guard. This unforeseen development underscores the challenge of accounting for exogenous factors and underscores the need for robust analytical methodologies to disentangle the effects of anti-corruption policies on household expenditures.¹²

Xi Jinping's anti-corruption campaign, notably featuring the Eight Regulations of Austerity, primarily targets government system employees. To discern the impact of these policies, I contrast expenditure patterns between families with at least one government system employee and those without. In theory, the effectiveness of these anti-corruption measures should manifest in the expenditure habits of the former group to a greater extent than the latter. Specifically, one would expect to observe a notable increase in relative spending by government households on goods and services previously obtained as gifts, while witnessing minimal changes in other expenditure categories. Furthermore, I conduct a falsification test, using private company-affiliated families as the treatment group, to assess the validity of the subsequent difference-in-differences estimation. The anti-corruption effect on "affected" and "non-affected" expenditures, as estimated through the difference-in-differences (DID) approach, is represented by Eq. (1):

$$\beta_3 = \left(Y_{post-2012}^T - Y_{pre-2012}^T \right) - \left(Y_{post-2012}^C - Y_{pre-2012}^C \right) \quad (1)$$

Where Y represents the households' expenditures on "affected" or "non-affected" items, T denotes the treatment group comprising

¹² The 18th National Congress discussed materials is from the Chinese Embassy in US and China.org.cn whose links are in the Appendix.

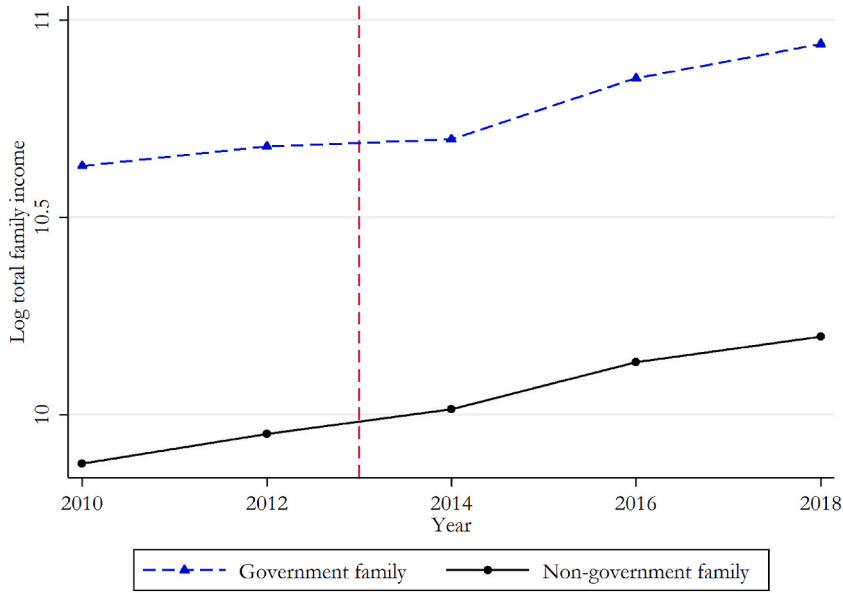


Fig. 3. Trend in Total Family Income from 2010 to 2018

Notes: The data represents the total family income surveyed by CFPS for the years 2010, 2012, 2014, 2016, and 2018.

Source: Author's calculations.

government-affiliated households, and C signifies the control group consisting of non-government-affiliated households. The expression within the first parentheses estimates the discrepancy in expenditure for government-affiliated households (treatment group), while the term within the second parentheses assesses the difference in expenditure for non-government-affiliated households (control group). The disparity between these two distinctions isolates the impact of the anti-corruption campaign on the expenditures of government-affiliated households.

Ultimately, to unveil the ramifications of the anti-corruption regulations, I juxtapose household expenditures across these two categories of goods before and after 2012, commencing from 2014—the initial year of data availability in the CFPS dataset post-2012—and contrast these disparities for both sets of families. Thus, the primary methodology for analyzing the anti-corruption campaign's effect on households' expenditures is as follows:

$$Y_{hh,t} = \alpha + \beta_1 gov_{hh} + \beta_2 post12 + \beta_3 gov_{hh} * post12 + X_{hh,t} + HH_t + \epsilon_{hh,t} \quad (2)$$

The household-level measurement, $Y_{hh,t}$, encompasses “affected expenditure,” “non-affected expenditure,” or “surveyed family total expenditure” across different years, where hh denotes households and t indicates different time periods. The variable gov_{hh} equals 1 if household hh belongs to the government-affiliated group and 0 if it pertains to the non-government-affiliated group. Similarly, the variable $post12$ is a time dummy variable set to 1 if the expenditure is observed after 2012 and 0 otherwise. Given that Xi Jinping's anti-corruption campaign commenced at the end of 2012, the policy's effective implementation period in the CFPS dataset commences in 2014, which is the earliest data available post-2012. Consequently, the time dummy variable $post12$ captures aggregate factors influencing both groups. The interaction term $gov_{hh} * post12$ represents the interaction between the treatment group dummy and the time dummy. The coefficient of this interaction term, β_3 , estimates the difference-in-differences effect, reflecting the impact of Xi Jinping's anti-corruption measures on expenditures.

$X_{hh,t}$ represents a vector encompassing household head demographic characteristics and household features, including gender, age, education level, and household size. To mitigate unobserved influences across households, household-level fixed effects are also controlled for. Given that household heads may change over different survey years and households may relocate, these fixed effects help account for such variations. The coefficient β_1 captures the disparity in spending between government and non-government households over the sampling period, while β_2 elucidates changes in average family expenditure across all households before and after the unexpected onset of Xi Jinping's anti-corruption campaign.

When comparing the results derived from equation (2) focusing on affected and subsequently non-affected expenditures, we anticipate observing more pronounced changes in affected expenditure among government workers only if the anti-corruption campaign is indeed prompting behavioral shifts or curtailing illicit benefits.¹³

¹³ I confirm the results from Eq. (2) with a triple differences model in Appendix Table A4.

4.2. Baseline results

Table 3 presents the baseline results for affected expenditures in panel A and non-affected expenditures in panel B. In both panels, column (1) demonstrates that government households significantly outspend non-government households in both affected and non-affected expenditure categories, with coefficients of 0.295 and 0.071, respectively. Moving to column (2) in both panels, the inclusion of an additional time dummy variable representing the anti-corruption policy shock reveals that both government and non-government households increase their spending on affected and non-affected expenditures following Xi Jinping's anti-corruption policy shock. Columns (3) and (4) build upon this by introducing the difference-in-differences (DID) interaction term and all control variables, encompassing family size, household head age, household head gender, family annual income, family annual savings, and household ethnicity dummy variables.

The coefficients of primary interest lie in the interaction terms, Gov*Policy, featured in columns (3) and (4), which exhibit a substantial and statistically significant positive effect in panel A. Conversely, in panel B, these coefficients are indistinguishable from zero both in terms of statistical significance and economic magnitude. This contrast suggests that following the anti-corruption campaign, government household expenditures on affected commodities rose by 12.5 percentage points more than those of non-government households. However, no similar increase is observed for non-affected expenditure categories, as evidenced by the results in panel B, column (4), which indicate that changes in government household spending mirror those of non-government households across non-affected expenditure groups. In essence, Xi Jinping's anti-corruption campaign appears to influence government household expenditures in affected expenditure categories while leaving their spending in non-affected categories largely unaffected. These findings underscore the effectiveness of Xi Jinping's anti-corruption policy in curbing expenditures associated with graft in everyday life.

4.3. Validity of the difference-in-differences method

To further validate the difference-in-differences regression specification, I employed four additional methods for robustness checks.

First, I used a placebo effectiveness year for the anti-corruption campaign, assuming it to be 2016 instead of the actual effective year of 2014, the first full year following the campaign's initiation. It is important to note that the anti-corruption campaign was implemented at the end of 2012, so 2012 should not be considered a post-campaign period. Due to the lack of detailed expenditure subcategories in the 2010 CFPS dataset, total expenditures were used. All other regressors remain consistent with those in Table 3, with the anti-corruption effective year set as 2016. If the control group is comparable and the difference-in-differences method is valid, the coefficient of the interaction term should be indistinguishable from zero. The insignificant coefficient estimate of 0.0242 for the DID interaction term in Column (2) of the left panel in Table 7 supports the validity of the DID approach.

Second, as a placebo test, I replicated the regression analysis from Table 3, designating households with private company employees as the treatment group and all other non-government employment households as the control group. The results, shown in the right panel of Table 7, reveal insignificant coefficients for the interaction term, indicating that the anti-corruption campaign did not affect the expenditures of households with private employment in either the affected or non-affected expenditure groups.

Third, I examined whether there was a differential trend in expenditures between government-affiliated and non-government-affiliated households prior to the anti-corruption campaign. The 2010 CFPS dataset differs from those of other years as it lacks detailed subcategories of expenditures used to classify affected and non-affected expenditure groups. To test the common trend assumption, I used the mean family annual total expenditures from 2010 to 2018, presented in Fig. 4. The results indicate a parallel trend before the anti-corruption campaign and a significant expenditure increase in government-affiliated households after 2014, suggesting a substantial impact of the campaign.

Fourth, I conducted a manipulation test to assess the unpredictability of the anti-corruption campaign, which was launched just 18 days after Xi Jinping assumed the presidency. The short time gap between the new presidency and the new policies implies a degree of randomness. In Figure A1, I examined the percentage of households with at least one government employee as a fraction of total households from 2010 to 2018, using the 2010 dataset for its employment information. Figure A1 shows an increase in the rate of government households following the December 2012 commencement of the anti-corruption campaign, peaking in 2016. This finding supports the notion of the campaign's randomness.

4.4. Robustness check

In this section, I investigate potential confounding factors and assess whether the impact of the anti-corruption campaign varies across different levels of government-affiliated households. One potential concern is changes in overall economic conditions. Therefore, I examined China's GDP, GDP growth rate, Chinese government revenue, and Chinese government annual final expenditures from 2010 to 2018, as presented in Appendix Table A3. To further assess the economic conditions and environment in China, and considering literature that discusses the sudden increase in money supply as a potential tool to "buy" votes in political contexts, I also examined M1 supply data from the official website of the People's Bank of China. Fig. 5.1 and 5.2, based on information from the People's Bank of China, indicate that the M1 supply increased steadily, with changes remaining stable on average. This analysis of M1 supply further reinforces the findings of this paper, aligning with insights from T. S. Aidt et al. (2020). The results indicate an increase in all these indexes, which may have amplified Chinese government revenues and expenditures, thereby expanding the potential pool of resources and opportunities for corruption among government system workers. However, the primary regression results, along with all robustness checks conducted in this study, reveal a different pattern: only lower-level government-affiliated households exhibited

Table 3
Effects of anti-corruption campaign on expenditures.

A: Affected Expenditure				
	(1)	(2)	(3)	(4)
Government	0.295*** (0.0307)	0.277*** (0.0302)	0.173*** (0.0551)	0.0132 (0.0619)
Policy		0.368*** (0.0307)	0.346*** (0.0320)	0.331*** (0.0332)
Gov * Policy			0.126** (0.0554)	0.125** (0.0593)
Controls	No	No	No	Yes
County FE	No	No	Yes	Yes
County Cluster	Yes	Yes	Yes	Yes
R-squared	0.574	0.581	0.581	0.627
N	43,547	43,547	43,547	39,329
B: Non-Affected Expenditure				
Government	0.071*** (0.0172)	0.056*** (0.0168)	0.050* (0.0275)	−0.028 (0.0296)
Policy		0.294*** (0.0180)	0.293*** (0.0193)	0.276*** (0.0200)
Gov*Policy			0.0073 (0.0258)	0.009 (0.0267)
Controls	No	No	No	Yes
County FE	No	No	Yes	Yes
County Cluster	Yes	Yes	Yes	Yes
R-squared	0.592	0.604	0.604	0.645
N	48,376	48,376	48,376	43,598

Notes: The dataset utilized in the above regressions spans the years 2012, 2014, 2016, and 2018. All standard errors are presented in parentheses. “Government” is a group indicator variable where 1 denotes the government group and 0 represents the non-government group. “Policy” is a time indicator variable corresponding to Xi Jinping’s anti-corruption campaign; 0 indicates the period before the campaign, and 1 indicates the period after the campaign. “Gov*Policy” represents the interaction term between the government group indicator and the policy time indicator. “Controls” include household head’s age, gender, family size, annual household income, annual household savings, residential location, and education level indicators. “County FE” refers to county-level fixed effects, accounting for unobserved heterogeneity at the county level. “County Cluster” indicates that standard errors are clustered at the county level to ensure robust statistical inference. Significance levels are denoted as follows: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

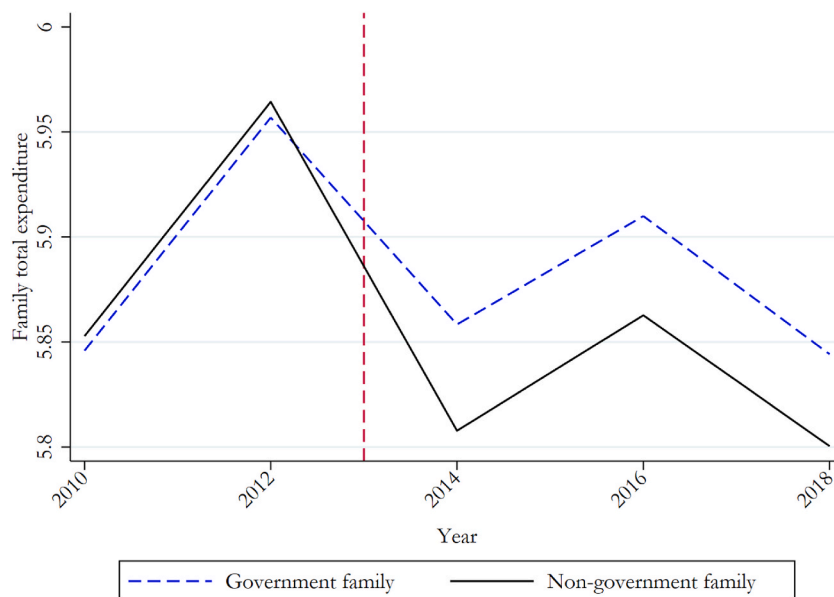


Fig. 4. Common Trend in Total Family Expenditure

Notes: The CFPS surveyed total family annual expenditure for the years 2010, 2012, 2014, 2016, and 2018.

Source: Author’s calculations.

an increase in their affected expenditures among households with government employees. Another concern is the uniformity of the anti-corruption campaign's influence across different levels of government household expenditures. To address this, I conducted a quintile regression analysis, classifying five ascending expenditure groups. The results are presented in Table 4. For government-affiliated households, regular expenditure survey responses are deemed more reliable than income survey responses. Thus, I used household expenditure as the classification standard to categorize ascending quintile groups for both affected and non-affected expenditures. In the affected expenditures panel, the DID coefficient is significantly positive in the lowest twenty percentile expenditure group, indicating a reduction in corrupt behavior of 32.3 %. However, the magnitude decreases to 11.6 % at a 10 % significance level in higher quintiles. From column (3) to column (4), the associated growth difference between government and non-government households before and after the anti-corruption campaign becomes insignificant.

While there may be a large effect among the richest households, it may not be precisely estimated. These findings suggest that the largest reduction in corrupt behavior may have occurred among the lowest-level (poorest) government system households, implying that they began to spend more of their own income on expenditures previously obtained free or at a discount. In middle-income and possibly higher-level expenditure groups, this trend may suggest that the anti-corruption effects through reduced illegal cash income channels could become more pronounced, or the anti-corruption campaign may have less impact on higher-level government households.

In the non-affected panel, the coefficients of the interaction term remain insignificantly small and negative across all columns (1) to (5). These consistent results, as observed in Table 3, indicate that the anti-corruption campaign did not significantly impact expenditure changes over the period for this group relative to non-governmental families, bolstering confidence in the causal interpretation of the findings.

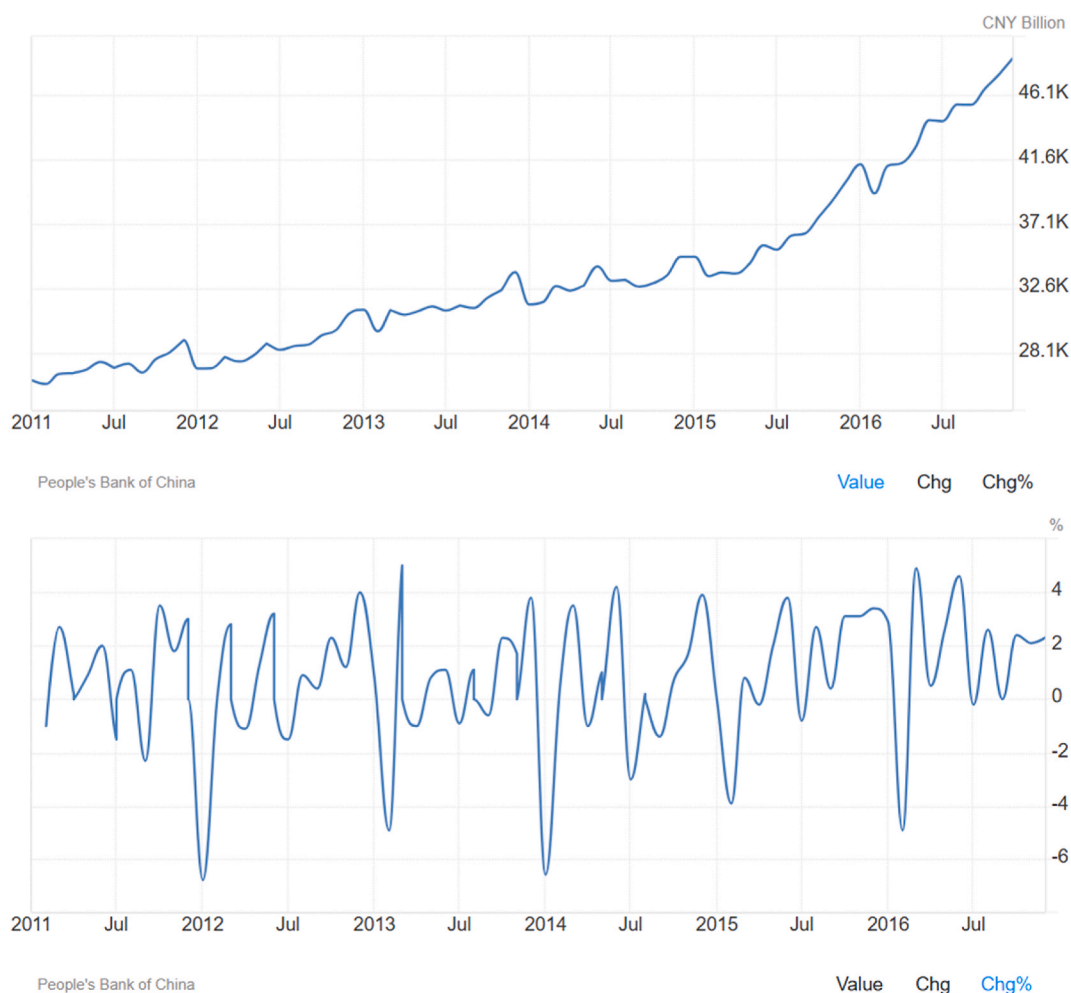


Fig. 5.1. Fig. 5.1 Trends in Total M1 Values (2011–2016)

Note: Data source: People's Bank of China.

Fig. 5.2 Trends in Percentage Changes of M1 Values (2011–2016)

Note: Data source: People's Bank of China.

Table 4
Quintile regression results.

A: Affected Expenditure					
	0–20 %	21 %–40 %	41 %–60 %	61 %–80 %	81 %–100 %
	(1)	(2)	(3)	(4)	(5)
Government	0.0005 (0.067)	0.101* (0.056)	0.192*** (0.050)	0.179*** (0.047)	−0.048 (0.218)
Policy	0.463*** (0.031)	0.472*** (0.026)	0.365*** (0.023)	0.173*** (0.022)	0.329*** (0.100)
Gov*Policy	0.323*** (0.074)	0.116* (0.062)	−0.0377 (0.056)	−0.0256 (0.052)	0.338 (0.241)
Controls	Yes	Yes	Yes	Yes	Yes
County FE	No	No	No	No	No
County Cluster	No	No	No	No	No
Pseudo R-squared	0.125	0.125	0.119	0.104	0.094
N	9800	9050	6950	5381	3800
B: Non-affected Expenditure					
Government	0.0904** (0.04)	0.118*** (0.03)	0.138*** (0.03)	0.151*** (0.04)	0.174 (0.117)
Policy	0.243*** (0.02)	0.224*** (0.02)	0.209*** (0.02)	0.171*** (0.02)	0.368*** (0.05)
Gov*Policy	0.0296 (0.04)	−0.00945 (0.04)	−0.0348 (0.04)	−0.0443 (0.04)	−0.0475 (0.129)
Controls	Yes	Yes	Yes	Yes	Yes
County FE	No	No	No	No	No
County Cluster	No	No	No	No	No
Pseudo R-squared	0.168	0.131	0.0956	0.0590	0.226
N	10,050	9905	7295	4335	3394

Notes: The data set refers to Table 3. For all variable's definitions including households fixed effects refer to Table 3. In these quintile regressions, I dropped top 5 % income group and top 5 % affected expenditures group. I did not drop any observations in non-affected expenditures regression. *p < 0.10. **p < 0.05, ***p < 0.01.

To provide further insights into how the anti-corruption campaign affected various government household expenditures, I divided the sample into urban and rural areas based on residential density. Given that higher-rank government organizations tend to cluster in urban areas while lower-rank ones are more prevalent in rural regions in China, the effectiveness of the anti-corruption campaign may vary across different government employees and households depending on their location (Hannum, 1999). Table 5, consistent with Table 4, suggests that the reallocation of expenditures predominantly occurred among lower-level employees, particularly in rural areas.

Another intriguing question is the extent to which ties with the Chinese Communist Party (CCP) can shield individuals from prosecution. Consequently, this paper classifies government households based on whether their members are also CCP members. Typically, CCP members within the government system hold higher-ranking positions. Even when non-CCP individuals are promoted to management levels, the government often encourages them to join the CCP (Shambaugh, 2008). Table 6 compares the results for households with CCP members to those without. The estimates suggest that expenditures in the affected category increased only among government households without CCP members following the anti-corruption campaign. In contrast, the anti-corruption program does not appear to affect consumption expenditures in CCP households. This result could imply that the anti-corruption campaign does not influence higher-level government households or that those with stronger ties to the party were more insulated from

Table 5
Expenditures in urban and rural.

	Rural		Urban	
	Affected	Non-Affected	Affected	Non-Affected
	(1)	(2)	(3)	(4)
Government	−0.0146 (0.110)	−0.0414 (0.0529)	0.0910 (0.0794)	−0.0190 (0.0424)
Policy	0.281*** (0.0446)	0.282*** (0.0255)	0.407*** (0.0524)	0.276*** (0.0302)
Gov * Policy	0.153* (0.079)	−0.0179 (0.0543)	0.0439 (0.0752)	0.00101 (0.0402)
Controls	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes
County Cluster	Yes	Yes	Yes	Yes
R-squared	0.602	0.643	0.675	0.686
N	19,565	22,021	19,764	21,577

Notes: The data set and all variable's definitions including all kinds of fixed effects refer to Table 3. *p < 0.10. **p < 0.05, ***p < 0.01.

prosecution and thus faced less incentive to change their behavior. Another potential explanation could be that households with government officials and CCP members were less involved in corrupt activities or had already reduced their corrupt behavior since the free-market reforms of 1978. However, according to reports on other anti-corruption campaigns and those involving convicted top Chinese government officials (as shown in Fig. 1), the first explanation seems more plausible (Deng et al., 2010).

Given that the CFPS dataset includes only one pre-treatment period prior to the anti-corruption campaign, I undertook several steps to bolster the credibility of the identification strategy. In addition to examining the parallel trends assumption in Fig. 4, the trends in total household income and expenditure in Figs. 3 and 2 respectively, and the general economic conditions presented in Table A3, I conducted a falsification test based on alternative timing of the policy intervention.

Specifically, I used the false start year 2014, as shown in Column (1) of the left panel in Table 7. As expected, the estimated effects on total household expenditure are statistically insignificant, aligning with the fact that both years fall within the post-intervention period and thus should not exhibit treatment effects.

Furthermore, I implemented an additional falsification test by redefining the treatment and control groups. Instead of designating government-affiliated households as the treatment group, I classified households affiliated with private companies as the treatment group and all other non-government-affiliated households as the control group. The results, presented in the right panel of Table 7, similarly show no statistically significant effects on total household expenditure, supporting the notion that the anti-corruption campaign primarily affected government-affiliated households and did not influence spending patterns among non-government-affiliated groups.

To further examine heterogeneous effects, I replace the baseline binary indicator for government-versus non-government-affiliated households with a continuous measure—the number of government employees within each household. The results are presented in Table 8. They indicate that households with more government-affiliated members experience significantly greater increases in both affected and non-affected expenditures, by 18 % and 9.1 % respectively, each significant at the 1 % level. The interaction term reveals a significant increase in affected expenditures at the 10 % level, but no significant effect for non-affected expenditures. These findings are consistent with the baseline results and support the conclusion that the anti-corruption campaign significantly increased spending on affected goods among government-affiliated households (see Table 3 and Table 9).

It has been suggested that Xi Jinping's anti-corruption campaign is merely a political tool to purge his enemies (Chen and Kung, 2019). To examine this hypothesis, I categorized the CFPS dataset into two subgroups: Xi Jinping's political power base provinces and provinces outside his power base. The promotion process within the Chinese government is unique; before a presidential candidate assumes office, they must work in various provinces to gain experience and enhance their capabilities. Xi Jinping has governed Fujian (1985–2002), Zhejiang (2002–2007), Shanghai (2007), and Beijing (2007–2012), so I classified these provinces as his political power base. I then ran Eq. (2) based on this classification and repeated the regression for the other provinces, as shown in Table 9.

The results suggest that the anti-corruption campaign had minimal impact, if any, in Xi Jinping's power base, but significantly influenced expenditure behaviors in other provinces. One possible reason is that government employees in Xi Jinping's power base are not as concerned about the anti-corruption campaign because they are loyalists. Conversely, the campaign increased government households' expenditures on affected items in other provinces. In the non-affected expenditure group, both subgroups exhibited minimal changes, lending credibility to the estimates.

Another reason could be that government officials in Xi Jinping's power base follow his directives more closely than those in other areas. Alternatively, Xi Jinping might have already implemented anti-corruption policies locally during his tenure in these provinces. Future research will need to explore this further. Based on the information in Fig. 1 and newspaper reports, the first explanation seems more plausible (Deng et al., 2010; Campbell, 2018; Shih, 2018). Based on later news regarding his removal of presidential term limits from the constitution, the former explanation appears more plausible.

Table 6
Expenditures in different treatment groups.

	Non-CCP and Government HH		CCP and Government HH	
	Affected	Non-affected	Affected	Non-affected
	(1)	(2)	(3)	(4)
Government	−0.0492 (0.0726)	−0.0745** (0.0329)	0.118 (0.0982)	0.0753 (0.0488)
Policy	0.336*** (0.0317)	0.275*** (0.0196)	0.332*** (0.0336)	0.280*** (0.0203)
Gov * Policy	0.159** (0.0693)	0.0231 (0.0333)	0.0538 (0.0888)	−0.0172 (0.0388)
Controls	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes
County Cluster	Yes	Yes	Yes	Yes
R-squared	0.627	0.645	0.641	0.656
N	39,329	43,598	34,699	38,704

Notes: “Non-CCP and Government HH” refers the treatment group, government households without CCP membership and “CCP and Government HH” refers another treatment group government households with CCP membership. The control group is identical to those used in all previous regressions. The data set and all variable's definitions including all kinds of fixed effects refer to Table 3. *p < 0.10, **p < 0.05, ***p < 0.01.

Table 7
Specification test.

	Time Specification Test		Alternative treatment control group	
	Placebo Test: Falsified Start Year = 2014	Placebo Test: Falsified Effect Year = 2016	Affected	Non-affected
	(1)	(2)	(3)	(4)
Government	0.0912*** (0.0246)	0.117*** (0.0171)	−0.0313 (0.480)	−0.0293 (0.195)
Policy Shock Year	−0.110*** (0.0221)	−0.00805 (0.0181)	0.255*** (0.0442)	0.262*** (0.0255)
Gov*Policy	−0.0484 (0.0299)	0.0242 (0.0258)	0.241 (0.491)	−0.00982 (0.193)
Controls	Yes	Yes	Yes	Yes
County FE	No	No	Yes	Yes
County Cluster	Yes	Yes	Yes	Yes
R-squared	0.310	0.308	0.693	0.713
N	32,240	32,240	23,843	26,937

Notes: Column (1) presents the falsification test using 2014 as the placebo start year of the anti-corruption campaign, while Column (2) uses 2016 as the falsification shock year. Columns (3) and (4) implement an alternative group falsification test, treating households affiliated with private companies as the treatment group and all other non-government-affiliated households as the control group. Robust standard errors are shown in parentheses. For definitions of all variables and details on fixed effects, see Table 4. *p < 0.10. **p < 0.05, ***p < 0.01.

Table 8
Heterogeneous effects by number of government employees in the household.

	Affected	Non-affected
	(1)	(2)
Num of Gov Employees	0.180*** (0.035)	0.091*** (0.020)
Policy	0.271*** (0.032)	0.218*** (0.019)
Num of Gov Employees*Policy	0.064* (0.037)	0.001 (0.020)
Controls	Yes	Yes
County Cluster	Yes	Yes
R-squared	0.685	0.707
N	0.219	0.250

Note: “Num of Gov Employees” indicates the number of government employees within each household. “Policy” is a time dummy variable indicating whether the observation falls before or after the anti-corruption campaign. The interaction term “Num of Gov Employees * Policy” captures the differential effect of household government employment in the post-policy period. For details on the dataset, variable definitions, and fixed effects specifications, refer to Table 3. Robust standard errors are reported in parentheses. *p < 0.10, **p < 0.05, ***p < 0.01.

Table 9
Expenditures in powerbase and non-powerbase.

	Non-powerbase provinces		Powerbase provinces	
	Affected	Non-affected	Affected	Non-affected
	(1)	(2)	(3)	(4)
Government	0.00944 (0.0646)	−0.0196 (0.0320)	0.0328 (0.233)	−0.107 (0.0807)
Policy Shock Year	0.325*** (0.0347)	0.277*** (0.0212)	0.395*** (0.0987)	0.271*** (0.0583)
Gov*Policy	0.129** (0.0624)	−0.00260 (0.0283)	0.0815 (0.206)	0.0605 (0.0816)
Controls	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes
County Cluster	Yes	Yes	Yes	Yes
R-squared	0.615	0.639	0.709	0.698
N	35,581	39,608	3748	3990

Notes: The data set in the regression spanning 2012, 2014, 2016, and 2018. All Standard errors in parentheses. For all variable’s definitions including all kinds of fixed effects refer to Table 3. In this regression, I dropped top 5 % income households for all households, because they are outliers. *p < 0.10. **p < 0.05, ***p < 0.01.

5. Conclusion

Xi Jinping's anti-corruption campaign is one of the most significant events in recent Chinese history, presenting an excellent opportunity for a natural experiment to analyze its effects. However, there are serious doubts about its effectiveness and whether it is a political maneuver to eliminate rival factions within the Communist Party rather than an economic reform. This paper addresses some of these questions, providing clarity on how the anti-corruption campaign affected various types of expenditures and which government-related families were most influenced.

To summarize, this paper examines Xi Jinping's anti-corruption campaign using a novel test based on expenditure patterns, classifying commodities into those susceptible and those not susceptible to corruption through the difference-in-differences (DID) method. In addition to the basic DID regression results, I also tested the campaign's effects on expenditures across different levels of government families, utilizing triple DID and analyzing various subcategories as robustness checks.

The campaign appears to have forced some government households to pay for products and services that they might have previously received for free. Interestingly, these effects are not uniformly distributed. Higher-level government households, those in the CCP, and those residing in Xi Jinping's traditional power base do not exhibit significant effects from the anti-corruption campaign. Instead, the changes are more pronounced among lower-level government households. These results suggest that the anti-corruption campaign primarily affected expenditures of lower-level government households, rather than those of higher-level government households or those with closer ties to the president. Future research may estimate the benefits of reduced corruption and further investigate these political economy characteristics.

While the CFPS dataset is currently the most reliable resource for understanding household expenditures in China, it does have certain limitations. Due to its start date, the CFPS dataset lacks at least two pre-anti-corruption campaign periods with detailed categorical expenditure data. Although this study utilized household total expenditures to establish the common trend assumption for the DID approach and discussed the societal and economic stability in China prior to the anti-corruption campaign—indicating no other events significantly influenced household expenditure behaviors—I acknowledge that having more granular categorical expenditure data would undoubtedly strengthen this research. Moving forward, I aim to access additional datasets and gather more detailed information on expenditures to further enhance the depth and robustness of this investigation.

Declaration of competing interest

“Assessing the Efficacy of China's Anti-Corruption Drive: Insights from Consumer Expenditure Patterns” is committed to integrity and fairness in the conduct of all its activities. Because this is a solo paper with no funding, there are no conflicts.

Appendix

1. Chinese government Eight Regulations of Austerity

The eight-point rules, issued by the Political Bureau of the CPC Central Committee in December 2012, aim to reduce bureaucracy, extravagance and undesirable work practices of Party members. With clauses focusing on various forms of corruption and unauthorized use of government cars, the rules have played a significant role in the country's anticorruption campaign.

1. Leaders must carry out in-depth inspections at the grassroots level so that they remain in contact with the people and understand the issues affecting society.
They must cut extravagance and do away with welcome banners, red carpets, floral arrangements and grand receptions for official visits.
2. Meetings and major events should be strictly regulated and simplified. Political Bureau members are not allowed to attend opening ceremonies, celebrations or seminars, unless they get approval from the CPC Central Committee. Official meetings should be economical and efficient, with no inconsequential or prolonged talk.
3. The number of official documents should be reduced to only those necessary.
4. Officials' visits abroad should be arranged only when necessary for foreign affairs, with smaller accompanying delegations.

On most occasions, there is no need for a reception by Chinese expatriates, institutions or students at the airport.

5. There should be fewer traffic controls when leaders travel by car to avoid unnecessary inconvenience to the public.
6. The media should consider the need, news value, and social impact of reporting on members of the Political Bureau, their work and their activities — the amount of time spent on such news items should be reduced and reports should be minimized in scope.
7. Leaders should not publish anything by themselves or issue any congratulatory letters in their own name unless an arrangement with the central authorities has been made.
8. Leaders must practice diligence and thrift, and strictly follow the relevant regulations on accommodation and cars.

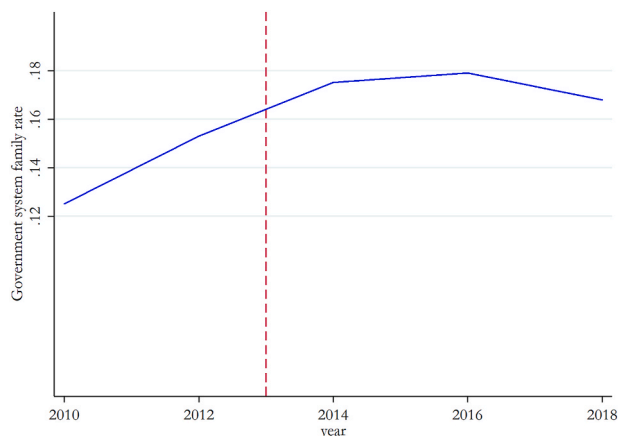


Fig. A1. Assessing the Randomness of Xi's Anti-Corruption Campaign

Notes: The government household rate is calculated as the number of government system households divided by the total number of households.

Source: Author's calculations using CFPS data.

Table A3
Economic condition in China

Year	RGDP/CAP	GDP growth rate	Revenue	Expenditure
2010	4550	10.636	11.231	12.846
2011	5323	9.551	11.239	13.217
2012	5832	7.860	11.139	13.475
2013	6343	7.769	10.956	13.589
2014	6754	7.300	15.874	13.375
2015	6990	6.905	16.110	14.036
2016	6892	6.737	15.764	14.386
2017	7355	6.757		14.522
2018	8038	6.567		14.676

Notes: RGDP/CAP represents GDP per capita based on 2010 US dollar values. Revenue indicates the Chinese government's annual revenue as a percentage of GDP, excluding grants. Expenditure refers to the government's annual final expenditure as a percentage of GDP.

Table A4
Triple DID Results

	Basic DID	Quantile regression				
	(1)	(2)	(3)	(4)	(5)	(6)
		20 %	40 %	60 %	80 %	100 %
Government	-0.147*** (0.0356)	0.0452 (0.0470)	0.0369 (0.0401)	0.0468 (0.0379)	0.0518 (0.0393)	0.165 (0.162)
Policy	0.221*** (0.0202)	0.261*** (0.0212)	0.231*** (0.0181)	0.219*** (0.0171)	0.170*** (0.0178)	0.495*** (0.0734)
Affected	-1.109*** (0.0356)	-1.545*** (0.0271)	-1.197*** (0.0231)	-0.881*** (0.0219)	-0.546*** (0.0227)	-2.306*** (0.0938)
Gov*Policy	-0.0185 (0.0324)	0.00782 (0.0528)	-0.0194 (0.0450)	-0.0620 (0.0425)	-0.0704 (0.0442)	-0.0474 (0.182)
Gov*Affected	0.197*** (0.0555)	0.0467 (0.0685)	0.227*** (0.0584)	0.279*** (0.0552)	0.261*** (0.0573)	-0.222 (0.237)
Policy*Affected	0.0503 (0.0329)	0.138*** (0.0308)	0.206*** (0.0262)	0.173*** (0.0248)	0.0352 (0.0258)	-0.207* (0.106)
Gov*Policy*Affected	0.147** (0.0572)	0.371*** (0.0772)	0.182*** (0.0659)	0.0570 (0.0623)	0.0243 (0.0647)	0.377 (0.267)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	No	No	No	No	No
County Cluster	Yes	No	No	No	No	No
Adjusted/Pseudo R2	0.597	0.247	0.194	0.150	0.100	0.271
N	55,122	9800	9050	6950	5381	3800

Notes: The data set in the regression and all variables' definitions refer to Tables 3 and 4. The dependent variables, expenditure, includes affected and non-affected consumption expenditures refer in equation (2). In the quantile regression panel, I dropped top 5 % income and top 5 % affected expenditures. I did not drop any observations in non-affected expenditures regression. *p < 0.10, **p < 0.05, ***p < 0.01.

Data availability

Data will be made available on request.

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